

Supplementary Material: DISEIL: Demonstration Distillation for Sample-Efficient Imitation Learning

Anonymous submission

Abstract

This document supports the main paper. It reports the complete ablation programme, the eighteen studies A1 to A18, in full and per setting; the hyperparameters and implementation details of every run; the prompts and environmental constraints; the derivations behind each rule; the per-round computational cost; and the qualitative record. Nothing here is required in order to understand the method.

A. Implementation and Hyperparameters

The implementation runs the four stages described in the DISEIL section of the main paper as a loop around an otherwise standard interactive imitation-learning harness. Every run stores its exact prompts and replies, so the material quoted in Sections K and L is a record rather than a reconstruction.

Table 1 lists every value the framework was run with. Three are worth separating. The context-set cap of three is defended from both sides by study A13 in Section G: two citations are worse and five buy nothing. The re-prescription limit of five bounds the feasibility loop, and the rounds that exhaust it are the fallback rounds whose rate study A6 reports. The threshold η is a quantile of the policy’s own training-loss distribution, recalibrated at each retrain, so the flagging rule tracks a policy that is improving.

Policies. The framework requires only that the policy expose a per-step loss, and it is instantiated with three policy classes to make that requirement visible. The four robot tasks use diffusion policies under both modalities (Chi et al. 2023), with an R3M encoder supplying the image branch (Nair et al. 2022). R3M supplies the policy’s visual representation and nothing else. It does not supply the clustering features, which are geometric in every run, state and image alike.

Retraining cadence. The cadence is a property of the task and not of the framework, and the reported runs do not use one cadence everywhere. On GridWorld the policy is re-trained from scratch after every round, so at $D = 1$ it is freshly fitted twenty times. On the four robot tasks, where a diffusion policy is expensive to fit, retraining runs once every fourth acquired demonstration, so the policy is refreshed five times over the budget while each of the twenty rounds still draws a fresh rollout pool. Both cadences hold for every arm, so no arm is advantaged by the schedule. The cadence enters

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the information-gain measurement of the main paper as well: on GridWorld the scoring policy is the policy of the round, and on the robot tasks it can be up to three demonstrations stale.

Round accounting. The seed counts are confirmed independently of the run logs. Clustered rounds plus skipped rounds total 180 for each GridWorld setting and 100 for each robot setting, which is seeds times B in both cases.

The starting-competence band. The initial demonstration set is excluded from the budget, and its size was not chosen freely. A policy’s starting success rate has to sit inside a band for the experiment to carry meaning. If the initial policy is too weak, its rollouts fail everywhere, every configuration is a failure, the failure set carries no structure for the descriptor to separate, and there is no allocation problem to solve. If the initial policy is too strong, the failure set is empty or nearly so, the budget has nothing to allocate, and every method converges to the same place. The band between those two conditions is the regime in which a fixed budget can be spent well or badly. Performance scales with the coverage of the demonstrations a policy is trained on, while demonstrations added within a setting already covered saturate (Lin et al. 2024), so the count is the lever that places a task inside the band.

The principle is implemented as a behaviour-cloning data-scaling sweep. A pool of expert demonstrations is collected, behaviour cloning is trained on each nested prefix of the pool, each prefix is evaluated on the frozen held-out set, and the prefix whose round-zero success rate is closest to a target near 50 per cent is selected. That sweep sets the $|\mathcal{D}_0|$ of Table 1 and the resulting round-zero success rates span 43.1 to 67.2 per cent, as recorded in the main results table. The count is set per task rather than per modality, so the two modalities of a task do not begin at the same success rate, and the image settings of Push-T and Door start slightly below the band, at 43.3 and 43.1 per cent. Lift begins the furthest above it, at 67.2 and 66.4, because the smallest prefix that trains a stable policy on that task already clears the band. That fact is the reason Lift sits at a ceiling in the main results and is uninformative about any mechanism.

B. Tasks, Experts and the Geometric Descriptor

Tasks. A setting is one task under one observation modality, and the evaluation covers five tasks under two modalities. The word *mode* is used only for a failure mode, which is a cluster of failures the framework discovers; an observation modality is never called a mode.

GridWorld 5x5 is a discrete navigation task on a five-by-five grid with three obstacle cells, in which an agent must reach a goal cell from a start cell. A* search and breadth-first search enter the task only as the feasibility and path-validity checker that decides whether a prescribed grid configuration admits an obstacle-free route from start to goal (Hart, Nilsson, and Raphael 1968; Cormen et al. 2022). They are never used as the expert and no policy is trained on their output. Push-T is a planar pushing task in which a manipulator must push a T-shaped block into a fixed goal pose. The task originates in the implicit-behaviour-cloning work that introduced it (Florence et al. 2021) and was popularised by the diffusion policy (Chi et al. 2023); the implementation used here is ManiSkill3’s PushT-v1 (Tao et al. 2024). ManiSkill3 is the third release of a benchmark line whose earlier releases carry a different task suite and do not contain Push-T (Mu et al. 2021; Gu et al. 2023), so the simulator and the task are cited separately. Lift, Wipe and Door are RoboSuite manipulation tasks on a UR5/UR5e arm (Zhu et al. 2020): lifting a cube from a table, wiping a randomised trail of dirt markers from a surface, and pulling a door open past a hinge threshold. The three RoboSuite tasks and Push-T supply the continuous-action half of the evaluation, and their reset distributions differ by an order of magnitude in width.

Experts. The expert differs by task, and each is named because the claim that the demonstrations are correct by construction depends on it. On GridWorld the expert is a human. On Lift, Wipe and Door the expert is a scripted oracle: an open-loop motion-planner routine on Lift, a closed-loop routine reading the hinge angle on Door, and a scripted wiping routine over the sampled marker path on Wipe. On Push-T the expert is a policy trained by proximal policy optimisation (Schulman et al. 2017), used without modification, so the Push-T expert is learned rather than scripted. It is an expert in the sense the framework requires, namely a demonstrator whose trajectories are the target the policy is fitted to, and it is not uniformly competent: the trained policy pushes in one rotational direction only, so configurations requiring the opposite rotation lie outside what it can demonstrate. Those configurations are excluded by the workspace constraints stored in the task’s constraint store, which is the store described in Section L.

The descriptor per task. The descriptor is the same six-dimensional vector in both modalities of a task. Table 2 gives its components. Position and orientation are the object’s, not the end-effector’s, on every task where an object exists, because the configuration that determines whether a failure recurs is the configuration of the thing being manipulated. Yaw enters through its sine and cosine so the wrap at $\pm\pi$ does not create a false distance between two nearly identical

orientations. The two columns of one task under the two modalities differ because an image policy fails in different places, not because the features differ.

C. Baselines and the Uniform-Random Control

Six comparison methods are run. Five of them are published interactive imitation-learning methods and share one skeleton: roll out the current policy, read a scalar signal, hand control to the expert when the signal crosses a threshold, aggregate the expert’s labels and retrain (Ross, Gordon, and Bagnell 2011). They differ only in the signal. SafeDagger learns a classifier that predicts when the policy is about to deviate from the expert (Zhang and Cho 2017). DropoutDagger reads the spread of a Monte-Carlo dropout ensemble of the learner’s action distribution (Gal and Ghahramani 2016; Menda, Driggs-Campbell, and Kochenderfer 2017). EnsembleDagger reads the variance of an explicit ensemble combined with an action-discrepancy term (Lakshminarayanan, Pritzel, and Blundell 2017; Menda, Driggs-Campbell, and Kochenderfer 2019). ThriftyDagger combines a novelty estimate with a learned risk estimate under a target switching rate (Hoque et al. 2021). Diff-Dagger uses a diffusion policy’s own per-step training loss as the uncertainty signal (Lee, Kang, and Kuo 2025) and is run on the robot tasks only, where the policy is a diffusion policy. Those five are the Dagger family and are labelled as such in every comparison. Diff-Dagger’s use of the per-step diffusion loss as an uncertainty signal is its own contribution; DISEIL uses that signal for failure localisation and also compares against it as a baseline, and both facts are stated wherever the signal appears.

The sixth comparison method, Stagger, is not a published system. It is a uniform-random control implemented in this project: each round corrects one uniformly chosen recorded failure, with no gate, no descriptor and no allocation. It carries no citation and it is never a member of the Dagger family. It is reported on GridWorld in the main results table, and its extension to the robot tasks is study A2 in Section F, where it answers the most damaging objection available against this work, which is that any failure replay would do.

D. Extended Derivations and Rationale

Each rule of the method is stated in the DISEIL section of the main paper. This section gives the reasoning behind the choices that the main paper states without room to argue.

Why the first crossing and not the peak. The flagged step is the first step at which the per-step loss exceeds η for K consecutive steps, and the peak is used only when no crossing occurs. The peak is the obvious alternative and it is the wrong one. In a failing episode the loss peaks late, because the policy’s error compounds, so an expert who takes over at the peak inherits a state that the policy has already ruined and has almost no episode left in which to correct it. The corrective demonstration then teaches recovery from a configuration the policy will rarely reach again. The first crossing is early, the state is less corrupted, and the demonstration begins where the policy first became unreliable rather than where it finished failing. The same argument is what makes

the descriptor informative: a descriptor computed at the peak would describe the wreckage rather than the cause.

The bound on the cluster sweep, and when it is skipped.

The cluster count is chosen by maximum mean silhouette over $k \in [2, k_{\max}]$ with $k_{\max} = \max(2, \min(6, N - 1))$. The lower bound of two is forced because a silhouette is undefined at $k = 1$. The cap at six is not a claim that six modes exist; it is a consequence of the sample size, since the failure sets are small and shrink over the budget, and a partition of a dozen failures into more than six groups is not a partition worth scoring. When fewer than four failures remain, the sweep is skipped entirely, each failure becomes its own singleton, and the round is allocated by the fallback rule. Study A15 in Section H reports how often that happens and study A17 reports when in the budget it happens.

The size constraint on the target. The target is the mode of highest mean peak loss among those whose size is within one member of the dominant mode’s. The constraint exists because mean peak loss is a mean over a set that can be very small. Without it, a single badly failed episode that lands far from every other failure becomes its own cluster, carries the highest mean peak loss by construction, and captures the round’s entire budget. The constraint keeps the target inside the bulk of the round’s failures, so the framework spends the round on a mode the policy is actually producing rather than on an outlier. It is the reason the prioritisation rule is not simply the maximum of $\bar{L}_C$.

Why feasibility is checked before the expert is called. A prescription is a request for a configuration of the world, and a language model asked for a configuration will sometimes ask for one the world cannot produce: an object outside the reachable set, a pose outside the spawn range, a grid layout with no path from start to goal. The store of environmental constraints is what makes such a request checkable. It is not a document store to be retrieved from in the manner of retrieval-augmented generation (Lewis et al. 2020; Edge et al. 2024); it is closer to the explicit, queryable environment and action knowledge of a robot knowledge base (Tenorth and Beetz 2013). A violation is returned to the model as feedback and a revised prescription is demanded until a feasible one is produced or the limit $J_{\max}$ is reached. A failed attempt consumes no budget, because the budget counts demonstrations collected and not prescriptions proposed. Propose-verify-revise against an external checker is a standard pattern (Liu et al. 2023; Chen et al. 2024b); what is new is not the checker but the object being verified, which is a request for a training demonstration that has not been collected rather than a plan to be executed.

The division of labour between the two model roles. The descriptor is computed from simulator state and the frames go to a vision-language model, and the two never mix. Vision-language models are competent at naming a cause when they are given structured evidence (Duan et al. 2025; Liu, Bahety, and Song 2023) and unreliable at metric and spatial reasoning from pixels alone (Chen et al. 2024a; Fu et al. 2024). The framework therefore asks them for the cause and computes the geometry itself. A mode’s name is the majority root cause

among its members, taken from a closed taxonomy stored in the task’s constraint store rather than invented by the model, because a partition returns integers and a method that reports a failure in mode 2 has said nothing.

The solvability screen. The second screen asks whether the current policy already solves the prescribed configuration and revises it if so, on the reasoning that a demonstration of something the policy can already do teaches nothing. Its nearest relatives are reverse curriculum generation (Florensa et al. 2017) and reset learning (Eysenbach et al. 2018). It is a design element only. It is not exercised in any run reported in this work, no number in either document is attributable to it, and it is not ablated anywhere, which is a gap in the study and is named as one in Section N.

Two derived quantities. The standard error throughout both documents is $\text{std}/\sqrt{n}$ with $n = 9$ on GridWorld and $n = 5$ on the robot tasks. *Margin retained* is the fraction of DISEIL’s advantage over the strongest baseline that survives an ablation, $(\text{ablated} - \text{best baseline})/(\text{full} - \text{best baseline})$, expressed as a percentage. It is reported alongside the raw damage because a component whose removal costs two points where the margin is three points is a different object from a component whose removal costs two points where the margin is ten. A value near 100 per cent means the component is decorative, a value near zero means it carries the result, and a negative value means the ablated system has fallen beneath the baseline it was built to beat.

E. The Ablation Programme: Scope and Conventions

The programme comprises eighteen studies. A1 to A13 remove or vary one component at a time. A14 to A18 are diagnostics: they measure a property of the running system rather than knock a component out of it.

The studies are run and reported on three settings, chosen to span the three policy classes and both observation modalities: GridWorld (image), where the policy is a convolutional network; Push-T (state), where it is a state diffusion policy; and Door (image), where it is an image diffusion policy. Every per-setting number below is one of those three, quoted in that order, and every aggregate is the mean over the three and is labelled as such.

The unit of analysis is the setting. Each study is reported as its three per-setting values, the sign they share and the mean over the three, with no aggregate figure standing in for them. The ablation programme is run only in settings with headroom, for the reason the main paper gives: where a framework and its baselines both sit at a ceiling, a null result cannot separate a component that does nothing from a component whose effect cannot be observed.

F. Knockouts, A1 to A8

The knockouts were run from the bottom of the system upwards and are reported in that order: first the two controls that establish what unstructured and unreasoning allocation can already do, then the components the method claims for itself. Table 3 is the full grid and Table 4 is the ladder in absolute

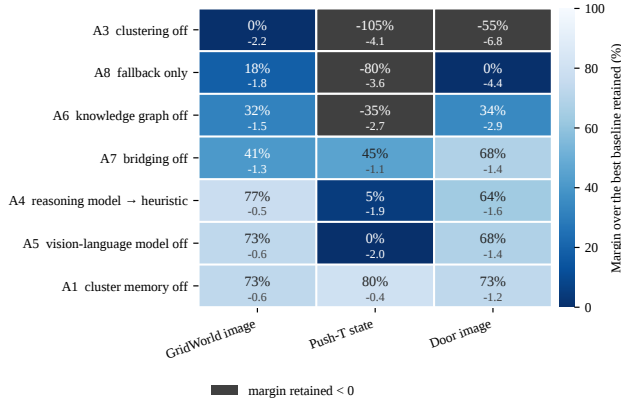


Figure 1: Cost of each knockout in points of final success rate against full DISEIL, averaged over the three ablation settings. The partition knockout is the most damaging; the remaining components cost gaps comparable to the seed standard error of the full run.

terms. Figure 2 plots the ladder and Figure 1 summarises the per-knockout cost.

A2, uniform-random allocation. The first rung corrects one uniformly chosen recorded failure each round, with no descriptor, no partition, no memory and no prescription model. On GridWorld this arm is the Stagger control of the main results table; on the robot settings it is a separate run. It reaches 89.1, 82.3 and 80.0 against DISEIL’s 91.3, 96.1 and 88.6. On the two robot settings it lands well below the strongest gated baseline. On GridWorld (image) it finishes level with the gated baselines and marginally above the best of them, and still below DISEIL.

Sampling failures uniformly reproduces the frequency of failure modes in the current policy’s failure set, so the dominant mode is corrected in proportion to how often it occurs and a rare, persistent mode is almost never touched inside twenty rounds. The premise of the whole framework is that the value of a demonstration is not proportional to the frequency of the corresponding failure, and A2 measures that premise directly. It also settles the obvious objection, that the advantage is nothing more than failure replay. On the robot settings random replay is worse than an uncertainty gate, and an uncertainty gate is worse than allocation; on the small GridWorld task the gates and random replay finish level, and allocation still leads.

A8, the deterministic fallback rule. When the prescription model cannot produce a feasible prescription after five attempts, DISEIL falls back on the nearest untried recorded failure in descriptor space. A8 promotes that rule to the entire method. It reaches 89.5, 92.5 and 84.2, costs 1.8, 3.6 and 4.4 points, and retains 18.2, -80.0 and 0.0 per cent of the margin. It beats the strongest baseline on GridWorld (image) only; on Push-T (state) it falls below it, and on Door (image) it lands exactly on it.

Nearest-untried is a spatial heuristic that implicitly spreads the budget, since a failure adjacent to one already corrected

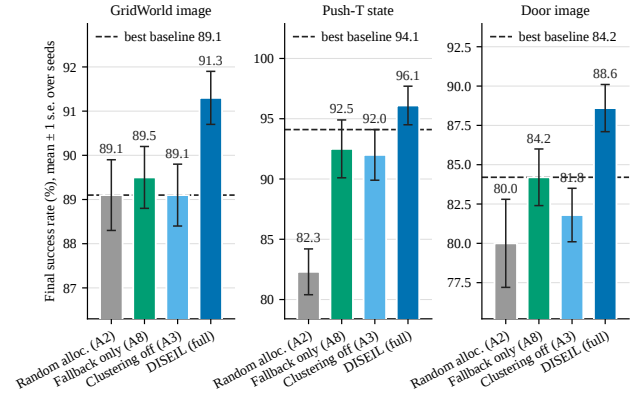


Figure 2: The allocation ladder on the three ablation settings. Bars are the mean final success rate over seeds with one standard error, for uniform-random allocation over recorded failures (A2), the deterministic nearest-untried fallback rule promoted to the whole method (A8), clustering removed in favour of greedy worst-loss selection (A3), and full DISEIL. The dashed line is the strongest baseline in that setting. Budget $B = 20$, one demonstration per round.

is less likely to be chosen than a distant one, and that is a crude version of the coverage pressure the partition and the memory supply deliberately. It captures a fraction of the margin at best, and it turns negative on Push-T (state), the setting whose strongest baseline is itself the strongest in the main results table. It cannot do better, because nearest-untried has no notion of a failure mode: it will select a chain of adjacent failures that all belong to the same mode, and it has no mechanism for concluding that a mode has been addressed. A8 is the floor against which every claim about the language and vision-language components is calibrated, alongside the baselines rather than in place of them.

A3, the clustering step, and the dissociation. The third rung removes the clustering step and targets, each round, the single failure with the highest peak per-step loss. The loss signal is kept and the failure-mode structure is removed. Success falls by 2.2, 4.1 and 6.8 points, a mean of 4.37, and the margin retained collapses to 0.0, -105.0 and -54.5 per cent, a mean of -53.2 . On Push-T (state) and Door (image) the ablated system falls beneath its own best baseline, 92.0 against 94.1 and 81.8 against 84.2, and on GridWorld (image) it lands exactly on it. Removing the clustering step erases the whole margin and on two of the three settings turns it negative, the largest damage of any knockout in the programme. Figure 3 plots the arm against its baselines.

What makes A3 the argument of the paper, and not one knockout among seven, is what happens to the information gain while the success rate is falling by those amounts. Per-demonstration information gain does not fall when clustering is removed. It rises on two of the three settings, by 0.02 and 0.16, and falls on the third by 0.13, for a mean change of $+0.02$. Greedy worst-loss selection is as good as DISEIL, or marginally better, at collecting demonstrations on which the current policy has high pre-retrain loss, which is unsurpris-

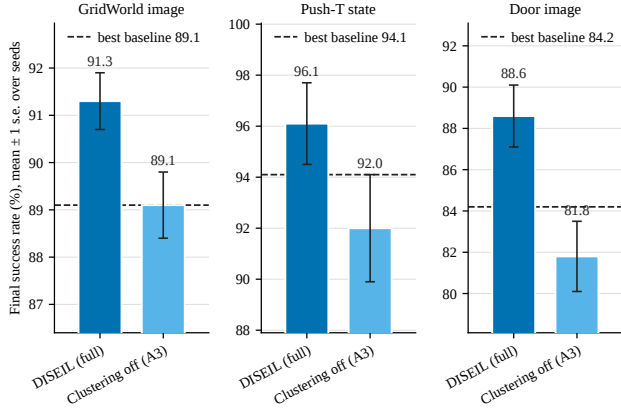


Figure 3: Success rate with clustering removed. Final success rate for full DISEIL and for greedy worst-loss selection (A3) on the three ablation settings, with the strongest baseline as a dashed line.

ing, since maximising that quantity is exactly what it does by construction, whereas DISEIL sacrifices some of it to spread the budget across failure modes. The two quantities move in opposite directions, and the dissociation is the finding.

The mechanism is straightforward once stated. Peak loss is a property of one failure trajectory. It is not a property of the *set* of failures the policy is still producing. Under greedy worst-loss, the highest-loss failure in round r very likely belongs to the same failure mode as the highest-loss failure in round $r - 1$, because the modes that generate the largest loss spikes are the ones the policy has learned least about, and one demonstration does not close a mode. The budget is spent repeatedly inside one region of the state space. Each demonstration in that stream is genuinely informative in isolation, which is why the gain does not fall, and each is largely redundant with the demonstration collected in the round before, which is why the success rate does not move. Figure 4 shows the per-setting distribution of the quantity that the main paper’s information-gain table summarises.

The main paper’s table reproduces Diff-DAGger alone as the comparison column, because Diff-DAGger’s gate selects on the same per-step diffusion loss the metric is computed from and is therefore the one baseline that could be expected to lead on it. The four other query gates and the uniform-random control are not reproduced as columns, and they add no argument the Diff-DAGger column does not already make: DISEIL’s mean gain is above every gate in all ten settings, GridWorld included, and above the control wherever the control was run.

A3 carries a limitation that is stated rather than deferred. It removes the descriptor and the memory along with the clustering, because those components have nothing to operate on once modes are gone, so it is a knockout of the allocation *stack* and not of the partition in isolation. The cleaner variant, which keeps the descriptor and replaces agglomerative clustering (Ward 1963) with a random partition of the failures into k groups, would separate grouping by geometry from

grouping at all. It was not run, and it is future work rather than a claim.

A6, the constraints the prescription is checked against. The prescription model proposes a configuration; constraints are retrieved from the task’s store, which holds workspace bounds, reachability, object and spawn ranges and controller limits as structured key-value knowledge; the proposal is checked against them; a violation is returned to the model as feedback and a revised proposal is requested until a feasible one is produced. A6 removes the store from both the vision-language and the reasoning prompts, so the loop has nothing to verify against.

The cost is 1.5, 2.7 and 2.9 points, a mean of 2.37, and 31.8, −35.0 and 34.1 per cent of the margin is retained, a mean of 10.3. A6 is the third most damaging knockout and it costs nearly twice what the prescription model itself is worth. The fallback rate rises to 27.1, 27.0 and 34.8 per cent of rounds, which means roughly five to seven of the twenty rounds are spent on a fallback correction rather than a prescribed one, a direct loss of a quarter to a third of the budget. Figure 5 shows both halves of the measurement.

The relation between the fallback rate and the damage is loose, and A8 is what explains it: a fallback round acquires the nearest untried failure, so a lost round still costs the difference between a prescribed correction and a fallback correction and not the whole round, which is why A6 costs two and a half points and not six. What the store buys follows from the same mechanism. It stops the prescription model from producing requests the environment cannot instantiate, and it does not make the model a better reasoner. Two limitations bound the claim. The per-task store is authored by hand and the study does not decompose it, so it cannot be said whether the workspace bounds alone would recover most of the damage or whether the failure taxonomy and the per-mode rules matter as well. And A6 knocks out only the first of the two screens; the solvability screen is not ablated anywhere.

A4 and A5, the two model roles. The *prescription model* is the text-only model of the main paper, and it fills two roles. In the first it assigns a root cause and a trajectory phase to each failure, once per failure, from the taxonomy stored in the task’s constraint store. In the second it turns the selected mode and its context set into the round’s demonstration request, once per round. A4 replaces the second of those with the deterministic rule “always target the dominant cluster representative”, operating on the same geometric clusters, so that the comparison isolates the prescription decision and not the partition; the root-cause labelling role is retained in A4 and still runs. A5 removes the vision-language model, leaving the prescription model with the geometric descriptor and the root-cause taxonomy. The two roles are removed together nowhere in the programme, and the phrase *reasoning stack* names both roles of the prescription model plus the vision-language call taken together. Figure 6 plots both arms.

A4 costs 0.5, 1.9 and 1.6 points and A5 costs 0.6, 2.0 and 1.4, so both average 1.33 points and both retain roughly half the margin, 48.6 per cent for A4 and 47.0 for A5. Every individual gap in both studies is comparable to the seed

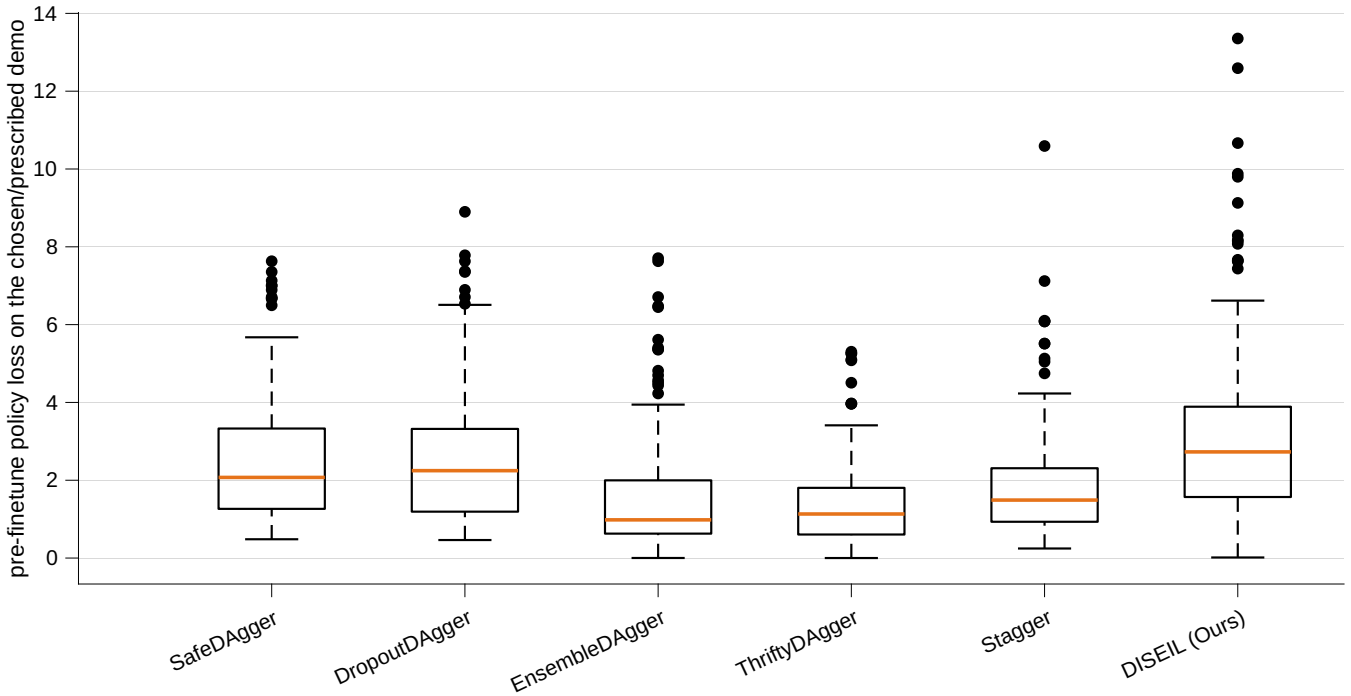


Figure 4: Per-demonstration information gain, the policy’s per-step loss on each newly acquired demonstration measured before retraining on it. The distributional view of the information-gain table in the main paper. Diff-Dagger is run on the robot tasks only.

standard error of the corresponding full run, and the two are close enough in magnitude that the study cannot rank them. Both are largest on Push-T (state) and smallest on GridWorld (image).

The explanation is structural. Clustering is geometric in every run, state and image alike, and it consumes no output from any foundation model. By the time either model is called, the decision that matters, which region of the failure distribution receives this round’s demonstration, has already been made by the descriptor and the memory. The prescription model chooses the form of the correction inside a region that was selected without it. A component acting downstream of the decisive step cannot produce a large effect, and the measurement agrees. Read the other way, the same fact is a practical result: a deployment that cannot afford the reasoning stack can delete it, keep the geometric clustering, the memory and the deterministic heuristic, and still beat every baseline on every setting.

Both knockouts carry a limitation that cuts against the framework’s interest. The A4 heuristic is a strong one. “Always target the dominant cluster representative” is itself an allocation rule, and it inherits the memory’s rotation, because the dominant cluster changes as the memory penalises recently corrected regions. A weaker heuristic would have produced a larger gap. The heuristic also cannot bridge, so A4 and A7 are not independent, and the study cannot separate what the prescription model buys as a reasoner from what it buys as the only component that can place a demonstration between two failures. A5, for its part, does not test whether

a better visual model would help more. Its largest gap is on Push-T (state), a state-modality setting and therefore counterintuitive, and the reading the architecture supports is that the frames are not there to compensate for a missing state vector. They are there to let the model see why the block ended where it did, and Push-T is the task on which one terminal geometry can be reached by several distinct failure processes, such as pushing on the wrong face, losing contact, or over-rotating. On Door, where the geometry of a failure largely determines its cause, the visual channel adds less.

A7, bridging placement. Bridging is the only part of the prescription that changes the environment configuration rather than selecting an episode, and it is the mechanism by which a prescription can be made easier than the failure it addresses. Disabling it costs 1.3, 1.1 and 1.4 points, a mean of 1.27, and retains 40.9, 45.0 and 68.2 per cent of the margin. That is proportionate to how often the arm is used: the bridged share of accepted prescriptions, measured as diagnostic A16 and shown in Figure 7, is 24, 28 and 21 per cent. A component used in a quarter of rounds cannot produce a large aggregate effect unless the rounds in which it is used are the decisive ones, and the three settings do not order the damage the way they order the share: the largest damage is on the setting with the middle share. What matters is which rounds bridge and not how many, and the coherent reading is that bridging pays when the target cluster lies far outside anything the current policy solves, so that a targeted demonstration would be a large distributional jump and a bridged one is a step the policy can absorb. That reading is not itself

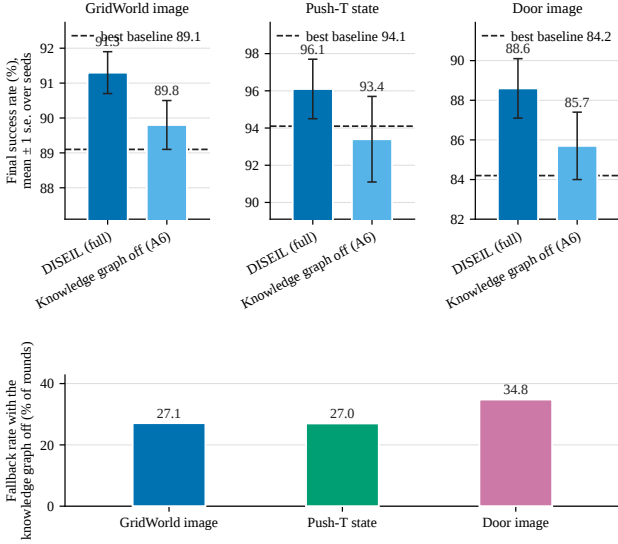


Figure 5: Grounding and feasibility. Top row: final success rate for full DISEIL and with the environmental constraints removed from the prompts (A6), against the strongest baseline. Bars are means over seeds with one standard error, including the A6 bar. Bottom row: the share of rounds that fall back to the deterministic rule when the constraints are removed.

tested here.

A1, the cluster memory. Switching the memory off, $\lambda = 0$, costs 0.6, 0.4 and 1.2 points, a mean of 0.73, and retains 72.7, 80.0 and 72.7 per cent of the margin, a mean of 75.1. It is the smallest effect of the seven knockouts and every individual gap is no larger than the seed standard error of the corresponding full run. The memory suppresses a cluster that has just received a demonstration, so the following round is pushed onto a different one, and A1 prices that suppression once the partition is already in place. The price is task-dependent: 0.4 points on Push-T (state) and 1.2 on Door (image), a factor of three across two of the three settings, so no single number describes what the memory is worth and the study does not supply one. The framework carries the memory as a configurable component of the loop, switched on per task, and not as a headline contribution. This study is the evidence for that description. What the framework has to defend is the partition the memory operates over, and A3 is where it is defended.

The kernel is also inert in most rounds. The candidate set of near-dominant modes is a singleton in 56 to 84 per cent of rounds on the settings with enough telemetry, and the dominant mode is then returned regardless of the penalty. The bandwidth is a single global constant while the tasks do not share a spatial scale. The memory is switched on in every run reported in the main paper and it is not drawn in the architecture figure.

The ordering. Ranked by mean damage, the seven knockouts are: clustering (-4.37 points, -53.2 per cent of the mar-

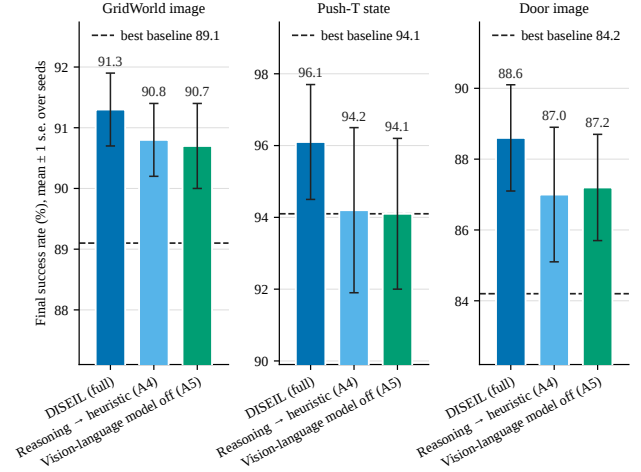


Figure 6: The prescription model and the vision-language model. Final success rate for full DISEIL, for the prescription model replaced by the dominant-representative heuristic (A4), and for the vision-language model removed (A5), with one standard error over seeds and the strongest baseline as a dashed line.

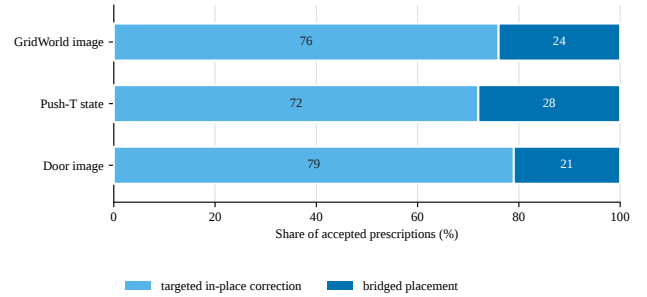


Figure 7: Targeted and bridged prescriptions (study A16). Share of accepted prescriptions on each of the three ablation settings that are a targeted in-place correction and that are a bridged placement.

gin retained), the fallback rule promoted to the whole method (-3.27 , -20.6), the constraint store (-2.37 , 10.3), the prescription model and the vision-language model (-1.33 each, 48.6 and 47.0), bridging placement (-1.27 , 51.4) and the cluster memory (-0.73 , 75.1). The ordering shaped the rest of the programme in two ways. The clustering step, which the framework treats as an implementation detail of a generic partition, is the component that carries the result. The cluster memory, which an earlier draft of the method advanced as one of its two contributions, is the least damaging of the seven and is reported as a configurable, task-dependent feature on that evidence.

G. Design Choices, A9 to A13

Given that allocation is the mechanism, the next family of studies asks whether *this* descriptor, *this* cluster count, *this*

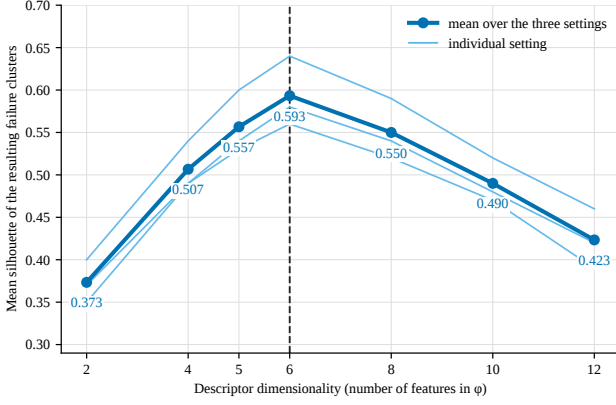


Figure 8: Mean silhouette of the failure clusters against the dimensionality of the geometric descriptor. One line per ablation setting and the mean over the three. The dashed vertical line marks the six-dimensional descriptor the framework uses.

context set and *this* budget are the right way to allocate. These studies were added after the knockouts, because until the knockouts had run there was no reason to interrogate the partition in such detail.

A10, the width of the descriptor. The descriptor is designed, not learned, which invites the suspicion that the feature set was fitted to the reported result. A10 answers by scoring the descriptor on what it is for, the separation of failure modes, using a criterion with no relationship to success rate: the mean silhouette of the resulting clusters (Rousseeuw 1987). Features are removed from and added to the descriptor and the silhouette is measured over every clustering round. If the feature set had been over-specified, the curve would be flat and the choice of six dimensions arbitrary.

The curve is a clean inverted U with a single interior maximum, and the six-dimensional descriptor is the highest-scoring variant in each of the three settings. Table 5 gives the mean and Figure 8 the per-setting curves. Below six dimensions the descriptor discards information that distinguishes modes, and the largest single step in the whole sweep, $+0.133$ in mean silhouette, is the step from two dimensions to four, which adds orientation. Position alone cannot separate a Push-T failure in which the block is in the right place at the wrong angle from one in which it is in the right place at the right angle and the pusher lost contact, so those failures collapse into one cluster and the cluster’s silhouette is poor. Adding task progress, worth $+0.050$, and contact distance, worth $+0.037$, each buy less, which is consistent with orientation being the dominant discriminator on these tasks.

Above six dimensions the fall is not an information loss, since every added feature carries some signal. It is geometric. As dimensionality rises, the ratio of the nearest to the farthest pairwise distance approaches one, all failures come to look equidistant, the agglomerative merge order becomes arbitrary, and the silhouette falls although nothing was removed. End-effector velocity adds two dimensions of mostly

noise to a distance computation over a few dozen points, and by twelve dimensions the joint-angle summary has added six more. The number of failures being clustered is small, forty-two in round 1 and two by round 20 on the instrumented setting, and distance concentration bites hardest at small sample sizes. The descriptor is small because the failure sets are small, and the two facts are linked by the geometry of the distance computation and not by a design preference.

A10 also settles a point of record. Clustering is geometric for every run, state modality and image modality alike, and the descriptor is the same six-dimensional vector in both. The limitation is the one silhouette cannot address: it scores geometric separation and not semantic correctness, and a descriptor could produce well-separated clusters that correspond to no distinct root causes at all. The purity diagnostic of Section H is the measurement that bears on that hole and it does not close it. A10 also shows only that this family of descriptors peaks at six dimensions. It says nothing about whether a learned descriptor would do better.

A9 and A13, the context set. The context set given to the prescription model contains three cited failure episodes chosen by three rules: the forced representative of the target cluster, the worst-peak-loss seed, and a farthest-point-sampling fill for diversity (Eldar et al. 1997). Written out, the set S referred to in the main paper’s Prioritise stage is built from the target mode C_{tgt} as

$$S_0 = \{\text{rep}(C_{\text{tgt}})\} \cup \left\{ \arg \max_{i \in C_{\text{tgt}}} \text{peak}_i \right\}, \quad S \leftarrow S_0,$$

$$S \leftarrow S \cup \left\{ \arg \max_{i \in C_{\text{tgt}} \setminus S} \min_{j \in S} \|\tilde{X}_i - \tilde{X}_j\|_2 \right\} \quad (1)$$

until $|S| = \min(\kappa, |C_{\text{tgt}}|)$,

with $\kappa = 3$ in every reported run and $\tilde{X}$ the standardised descriptors. A9 removes each rule in turn and adds a floor control of three episodes drawn at random from the cluster, with the target cluster fixed by the memory in every arm so that only the composition of the set varies. All figures in this paragraph are means over the three ablation settings against a full-system reference of 92.0, and they are collected in Table 6.

The three single-rule removals cost almost the same, so on this run the study cannot rank the three rules against one another. Each still hurts, and each for a reason the others do not cover. The forced representative guarantees the prescription model sees an example of the mode it has been instructed to fix, without which the cited episodes can all come from the cluster’s periphery. The diversity fill keeps a context set of three episodes from all crowding near the loss peak, where it would describe the mode narrowly and draw a narrow correction. The worst-loss seed sharpens the description of the mode. Random selection of three episodes costs 3.6 points, more than any single-rule removal, so the three rules together beat an unstructured draw and are complementary rather than redundant.

The magnitudes discipline the claim. The whole spread from the full context set to random selection is 3.6 points, larger than the constraint store and still less than the clustering. A13 sharpens the point by varying the *number* of cited

episodes jointly with the selection rule, and it contains the one comparison that could have deleted a component of the method: citing the top three episodes by plain peak-loss rank, against the three-rule construction, at the same number of citations. The gap is 2.13 points. It is a real gap in the paired sense, since the full construction wins on each of the three settings, so the farthest-point construction earns its place rather than barely surviving. Citing only two episodes costs 1.93 points, so fewer citations are worse; citing five gives no measurable gain over three; and citing every failure in the target cluster only raises the prompt length without adding discriminative detail, since the early rounds carry about forty failures and a context set of that size buries the target mode in detail the model cannot weigh. The cap at three is therefore justified from both sides. The single-citation arm was not run, and the reason is that it is confounded by construction: bridging requires at least two cited failures in order to define a placement between a failing region and a solved one. It is not a null result and must not be reported as one.

A12, the cluster count. The cluster count is chosen per round by maximum mean silhouette, which is standard practice, is used as such, and is not claimed as a contribution (Rousseeuw 1987; Pedregosa et al. 2011). A12 replaces the adaptive choice with a fixed k . Silhouette selection wins on each of the three settings, and it beats the best fixed alternative, averaged over the three, by 4.1 points: a fixed $k = 2$ costs 4.1 points against the adaptive rule, $k = 3$ costs 4.6, $k = 4$ costs 7.4 and $k = 5$ costs 6.7. The effect is well outside the seed standard error of the three settings, which runs from 0.6 to 1.6 points, so the adaptive rule is defended by the size of its effect and not only by the consistency of its sign.

No single fixed k is best across the three settings either: the best fixed value is $k = 3$ on GridWorld (image), $k = 2$ on Push-T (state) and $k = 5$ on Door (image), which is the point. Too few clusters merges distinct failure modes, so the memory penalises a merged cluster and suppresses correction of a mode that was never addressed. Too many splits one mode across several clusters, so the memory cannot recognise that the mode has been corrected and rotation is diluted across fragments of the same region. The right number varies by setting and by round, and only the adaptive rule tracks it. Figure 9 plots A9, A12 and A13 together.

A11, the budget. The framework is claimed to operate under any fixed budget, with $B = 20$ as the validated instance, and A11 is the experiment that supports or refutes the claim. The margin is far larger at the smallest budget than at the two larger ones: it averages +9.07 points at $B = 10$, +2.87 at $B = 20$ and +2.83 at $B = 40$, with per-setting values of +5.7, +9.0 and +12.5 at the smallest budget and +1.5, +3.2 and +3.8 at the largest. The decline is clean on GridWorld (image) and Door (image); on Push-T (state) the margin is +9.0 at $B = 10$ and then sits near two to three points at the two larger budgets, so the trend is not strictly monotonic there.

The margin shrinks because the baseline catches up and not because DISEIL degrades, and Figure 10 is the evidence. DISEIL’s own success rate rises with the budget on every one of the three settings, from 86.8 to 94.0 on GridWorld

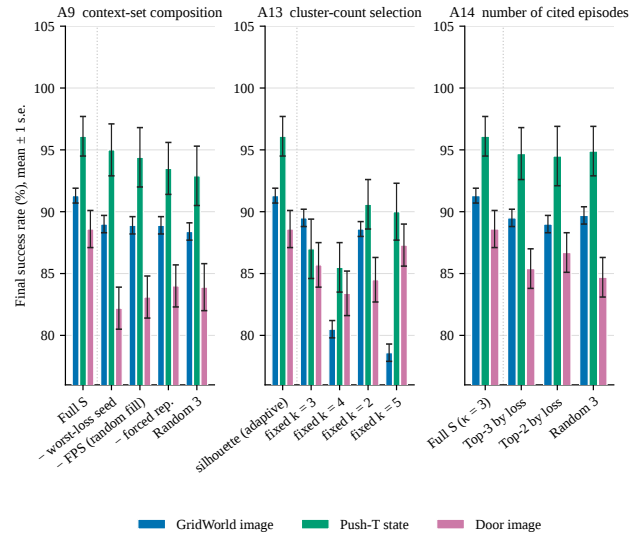


Figure 9: Three ablations of the machinery inside individual steps, with arms ordered by effect. Left: the composition of the context set (A9). Middle: silhouette-based selection of the cluster count against fixed alternatives (A12). Right: the number of cited episodes and the selection rule (A13). Bars are means over seeds with one standard error, on the three ablation settings.

(image), from 87.9 to 97.7 on Push-T (state) and from 82.3 to 99.5 on Door (image), while the strongest baseline rises faster from a lower start. Allocation buys the *rate* at which the failure distribution is covered and not the asymptote, which is why a large budget closes the gap: with enough draws, even a poorly allocated stream eventually covers the failure distribution, because coverage is a coupon-collector problem and the collector wins if it draws long enough.

One claim the sweep was expected to support does not hold, and it is retracted here. The study was designed around the headline that DISEIL at $B = 10$ matches the strongest baseline at $B = 20$, which would be the same policy for half the expert labour. It does not. On the three ablation settings, DISEIL at $B = 10$ reaches 86.8 against the baseline’s 89.1 at $B = 20$ on GridWorld (image), 87.9 against 94.1 on Push-T (state) and 82.3 against 84.2 on Door (image), so it falls short on every one. The halved-labour claim is false and appears nowhere in either document. What survives is the claim the data do support, which is the one that matters: the advantage of allocation grows as the budget shrinks, and the method stops paying when demonstrations stop being scarce.

H. Diagnostics, A14 to A17

The last family of studies removes nothing. It asks whether the clusters the framework discovers mean what the method says they mean, and whether the machinery that discovers them stays active for as long as the budget runs. Both answers bound the claims above rather than adding to them.

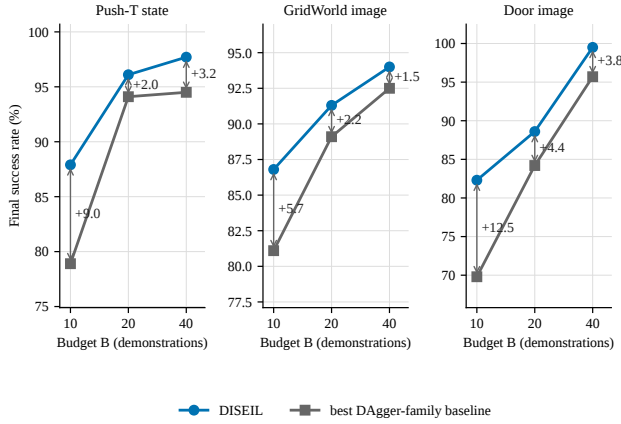


Figure 10: Final success rate against the budget B (study A11). DISEIL against the strongest baseline at $B = 10, 20$ and 40 , on the three ablation settings, with the margin printed between the two series.

A14, and the limit of a geometric descriptor. A10 shows that the descriptor produces well-separated clusters. A well-separated cluster need not correspond to a single root cause, and A14 is the check that stops the failure-mode claim from being a statement about geometry alone. Purity runs from 0.84 to 0.91 across the three settings and the number of distinct root causes per cluster runs from 1.35 to 1.86. Table 7 gives the grid.

The ordering is the informative part and it is the same ordering on both columns. Push-T (state) has the best silhouette, 0.64, and the highest purity, 0.91. Door (image) has the lowest silhouette, 0.56, the lowest purity, 0.84, and the most root causes per cluster, 1.86. Geometric separation and semantic purity rise and fall together on these three settings, so A10 and A14 are not independent audits of the descriptor: a descriptor that separates well geometrically also tends to produce clusters that are semantically clean, and neither measurement checks the other. What A14 establishes is the qualification and not an independent guarantee. The descriptor separates failures by where and how they occur, and it recovers root cause only to the extent that configuration determines cause. Where several causes share one end-effector position, as they do wherever a cluster carries close to two distinct root causes, geometry cannot tell them apart.

The circularity in the measurement is admitted. Purity is scored against the reasoning model’s own root-cause labels, so it records agreement between two components of the same system. There is no human-labelled root-cause set, and there should be, because purity and silhouette covary and a human-labelled set is the only measurement that would audit either of them from outside.

A15, the distribution of the cluster count. A15 shows that the adaptivity is not a disguised constant. Pooled over the 308 clustered rounds of the three ablation settings, $k = 3$ is the most frequently selected count at 26.3 per cent, with $k = 4$ at 23.4, $k = 5$ at 21.4, $k = 2$ at 14.9 and $k = 6$ at 14.0.

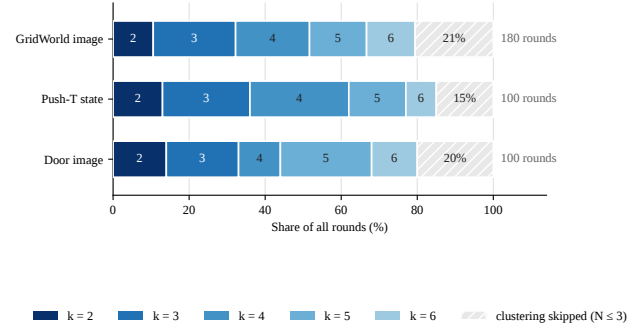


Figure 11: Cluster count selected per round (study A15). Share of all rounds selecting each cluster count k , with the rounds that skip clustering entirely, because fewer than four failures remain, shown hatched.

No value from two to six falls below 14 per cent. The claim the framework may make is that the number of discovered failure modes varies by round and is most often three or four, and not that there are three failure modes.

A15 also carries a seam. Twenty-one per cent of GridWorld (image) rounds, 15 per cent of Push-T (state) rounds and 20 per cent of Door (image) rounds never cluster at all, because fewer than four failures remain, and in those rounds each failure becomes its own cluster and the budget is allocated by the fallback rule. Figure 11 shows the distribution with those rounds marked.

A16, the bridged share, and a discrepancy of record. The share of accepted prescriptions that are bridged is 24 per cent on GridWorld (image), 28 on Push-T (state) and 21 on Door (image), plotted in Figure 7 alongside study A7.

One discrepancy in the record is stated rather than smoothed over. The method’s precondition for bridging is pose randomisation, from which it follows that bridging should be inapplicable on GridWorld, which is a discrete grid. The prescription logs disagree: 24 per cent of accepted prescriptions on GridWorld (image) are marked bridged, and A7 records a measurable effect there. Either the implemented mechanism is broader than the method text claims, or the runs did something the method text did not intend. The data are the source of truth, so bridging is reported as active on that setting, and the resolution of the discrepancy, by inspection of the prescription logs, is an outstanding item listed in Section N.

A17, failures per round. Failures per round fall from forty-two to two over the budget, halving by round 8 and falling by an order of magnitude by round 17. The decline is the system working as intended, and it also bounds the system’s own mechanism. Clustering forty failures into three or four failure modes is a meaningful operation. Clustering five failures into three is barely one. On the instrumented setting the descriptor, the clustering and the memory do their work through round 17, and the last three rounds run the fallback rule on a handful of remaining failures. A15 confirms the pattern on the three ablation settings, where 15 to 21 per cent

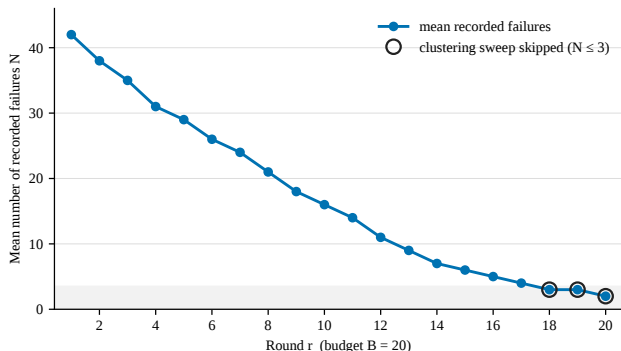


Figure 12: Mean number of recorded failures per round over the budget (study A17). The setting is Push-T (image), averaged over five seeds. Below four remaining failures the clustering sweep is skipped and each failure becomes its own cluster; the shaded band marks that region.

of all rounds skip clustering altogether. Figure 12 plots the curve.

This diagnostic is instrumented on one setting, Push-T (image), the only setting for which the failure count was logged per round. It is a diagnostic of the loop rather than an arm of an ablation, and the instrumentation should be extended to the three ablation settings before the front-loading behaviour is claimed as general, because the shape may differ where the initial success rate is higher. That is an outstanding item.

Read alongside the budget sweep, where the margin more than triples as the budget is halved, A17 says that DISEIL front-loads the value of a small budget, and the two facts are consistent: the budget’s marginal value is highest early and the machinery is most active early. It also names an extension that is not tested here and is not claimed, which is to stop the reasoning stack once the failure count drops below the clustering threshold and spend the remaining rounds on the fallback, saving most of the per-round reasoning cost at no measured loss.

I. Computational Cost, A18

The per-round cost of the reasoning pipeline is a limitation of this framework and it is the first quantity a reader will ask for once the success rates are accepted. Study A18 measures it directly, so that the limitation is stated with a number attached.

The measurement. Each of five settings runs a matched pair of runs, DISEIL against SafeDagger, on the same task, the same observation modality and the same hardware, and the wall-clock and token cost of every round is read out of each run’s own telemetry. SafeDagger is the baseline arm throughout. Two protocols are reported and they must not be mixed. Protocol P1 reports the first round of each run. It is the only protocol available in all five settings, and the first round holds the weakest policy and therefore the most failures and the most model calls, so P1 is an upper bound on steady-state cost rather than an average of it. Protocol P5 reports the

mean and the sample standard deviation over the language-model-active rounds of the runs that carried a longer budget, matched arm for arm against the baseline’s same-indexed rounds. The spread quoted under P5 is the round-to-round spread inside a run. A P1 row and a P5 row measure different objects and no reader should read one against the other.

What the measurement says. A round’s wall-clock is dominated by the from-scratch policy retrain and the held-out evaluation, and both arms pay all of it. Under P1 that shared cost is 783 to 1,491 seconds per round on the three RoboSuite settings, against a reasoning add-on of 270 to 700 seconds, and under P5 it is 548 to 1,333 seconds against an add-on of 233 to 656. The ratio of a DISEIL round to a baseline round inherits the large shared denominator and is diluted by it: across the two protocols it runs from 1.13 to 2.75, and a ratio near one is not evidence that the reasoning pipeline is cheap. It is evidence that the retrain is expensive. The quantity that characterises the pipeline is the add-on, the seconds and the tokens the baseline never spends, which is 63 seconds per round on GridWorld and 1,232 on Push-T, at 9,560 to 82,116 tokens per round. Both numbers belong in the record and neither on its own is honest.

Push-T is the outlier and its reasoning is not the reason. Its shared cost is the second smallest in the sweep and its baseline arm is unusually cheap, because SafeDagger’s loop runs until the intervention budget is spent and therefore halts after its first intervened episode, which is one episode and 6.3 seconds, while DISEIL screens a fixed sixty. At a longer budget the baseline would roll out many episodes per retrain and the ratio would fall. The token draw is the second reason: Push-T issues twenty language-model calls against Door’s seven. GridWorld sits at the other end in seconds and not in tokens. Its whole round is 118 seconds, so 63 seconds of reasoning is more than double a 55-second baseline round, and it still draws 9,735 tokens, of which the constraint block is 54 per cent of the prompt budget, the largest share of any setting. A cheap task in seconds is not a cheap task in tokens.

What this means for the framework. The language and vision-language models run only at demonstration-selection time. They are never in the control loop and they do not run at execution, so their cost is paid once per round, and on the three RoboSuite settings it is amortised over a retrain and an evaluation that together cost more than the reasoning does. The resource being traded is model inference against expert demonstration time, and on the settings measured here a round buys one demonstration for an extra 63 to 1,232 seconds of inference. Whether that trade is worth making depends on what a demonstration costs, which in simulation is nothing and outside simulation is a person’s time. The limitation is quantified and it is not removed. The pipeline costs real seconds and real tokens per round, the cost is highest exactly where the failure count is highest, and the graceful degradation that A4 measures, in which a deterministic heuristic over the same geometric clusters costs 1.33 points and still beats every baseline in every setting, is available to anyone for whom the trade does not clear.

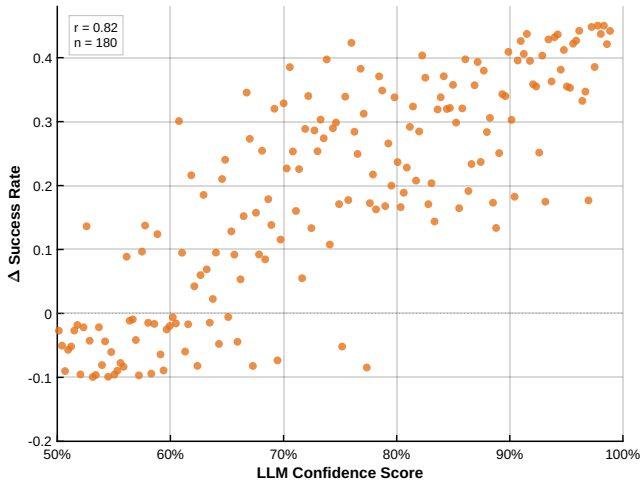


Figure 13: Reported prescription confidence against realised improvement. The confidence the prescription model reports at prescription time, against the change in the policy’s success rate on the round-level rollout evaluation that the resulting demonstration produced. GridWorld (image), 180 prescriptions, Pearson $r = 0.82$.

J. Prescription Confidence

At the moment it issues a prescription, the prescription model also emits an integer confidence between 0 and 100 together with a one-line rationale, reporting how likely it believes the resulting demonstration is to improve the policy. That number was added to the prompt as an instrument, to find out whether the model has any usable forecast of the value of the round it is about to spend. It is scored against ΔSR , the change in the policy’s success rate on the round-level rollout evaluation across that round.

The Pearson correlation between the reported confidence and the realised ΔSR runs from 0.82 to 0.89 across the ten settings. Quoting each task as state then image: GridWorld 0.88 and 0.82, Push-T 0.87 and 0.88, Lift 0.88 and 0.89, Wipe 0.82 and 0.86, Door 0.83 and 0.82. Figure 13 shows the GridWorld (image) scatter, where $r = 0.82$ over 180 prescriptions, the full count that nine seeds at a budget of twenty supply.

What makes this number usable, and not a rationalisation after the fact, is the order in which the two quantities become available. The confidence is reported blind, at prescription time. At that moment the demonstration has not been collected, the expert has not been called, the policy has not been retrained, and the re-rollout that produces ΔSR has not been run. The success-rate signal arrives only after all three of those steps have completed, by which point the round’s unit of budget has already been spent. The model is therefore forecasting an outcome it cannot observe, and a correlation of 0.82 to 0.89 is the accuracy of that forecast rather than a description of an outcome the model was shown.

The number carries two limits, and they are the reason it is reported here rather than as a result of the paper. The correlation is measured on DISEIL runs only, so it says nothing about whether a baseline’s gate signal would predict its own

ΔSR equally well. And no experiment gates on the confidence: nothing is skipped, deferred or re-prescribed on the basis of a low score, so the correlation is an observed property of the system and not yet a mechanism inside it. Turning it into one, by declining to spend a demonstration on a prescription the model itself does not believe in, has not been run.

K. Prompts and the Failure Taxonomy

Every run stores its exact prompts and replies. The material below is a record. The three model calls are the perception call, the reasoning call and the prescription call.

The perception call takes a system instruction and three frames of a cited failure: the start of the episode, the flagged step, and the final step.

The model’s reply on that instance reads: *“The failure occurs during the sweep at the beginning of the trajectory, around step $t^*=146$. At peak loss, the robot’s wiping pad is positioned at the start of the dirt trail, but the pad is tilted and not fully in contact with the table surface. The gripper is misaligned, causing the pad to lift slightly and only partially engage the dirt. This lack of stable contact prevents effective wiping, leaving the initial segment of the trail untouched.”*

The reasoning call is constrained to the vocabulary of the task’s constraint store and to strict JSON. The closed taxonomy is visible in the prompt: the root cause is one of six categories and the phase one of five, and the model’s job is assignment rather than invention.

Its reply on the same failure is the grounding requirement in miniature, because the rationale cites both the visual evidence and a named node of the store.

```
{
  "root_cause": "contact_instability",
  "phase": "transport",
  "rationale": "The wiping pad tilts and lifts during the sweep (transport phase) as described in the VLM evidence, causing intermittent contact with the table surface and aligning with the KAG-reasoned 'contact_instability' failure due to loss of downward pressure/alignment."
}
```

The prescription call states the budget rule to the model, offers the two arms of the prescription, and requires a confidence line.

L. Environmental Constraints: Representative Examples

The store of environmental constraints for a task is a JSON document with a fixed schema: metadata, typed nodes with key-value properties, relations between them, and a block of reasoning implications, one per failure mode plus a workspace constraint and a non-emptiness rule. A renderer turns the document into the text block injected into the reasoning and prescription prompts. The store is authored once per task, and its constraints are measurements of the environment rather than opinions about it.

SYSTEM: You are analysing a robot manipulation failure from rendered frames. Be concrete and spatial; describe what you actually see, not generic advice.

USER: You are analysing a robot manipulation failure. The attached frames are, in order: start, high_loss, end (the peak-loss frame is the policy's most-uncertain step, $t^*=146$).
Task: A UR5e robot with a wiping pad (no fingers) must WIPE a trail of dirt markers off the table by pressing down and sweeping along the dirt path until all markers are cleared (coverage).
Describe what went wrong. Focus on: where in the trajectory the failure occurs, the robot/gripper configuration at peak loss, and what object or contact state caused it. ~120 words, concrete and spatial.
[start frame] [high_loss frame] [end frame]

Figure 14: The perception call, verbatim from a Wipe run. One defect in the template is visible and is recorded rather than corrected: the template names the anchor the “peak-loss frame”, while the frame actually passed is the first-crossing step. The wording is stale and the frame is correct.

SYSTEM: You are a robot-manipulation failure analyst. Classify the root cause and trajectory phase using ONLY the provided categories and the KAG facts. Output strict JSON, no prose, no code fences.

USER: TASK: {task_description}
{kag_text}
VLM FAILURE DESCRIPTION (the only visual evidence): {vlm_report}
Identify the root cause category and the trajectory phase where the failure occurred.
root_cause in [grasp_failure, approach_failure, placement_error, contact_instability, pose_mismatch, timeout]
phase in [pre_grasp, grasp, transport, placement, insertion]
Output ONLY this JSON:
{"root_cause": "<one of the categories>",
"phase": "<one of the phases>",
"rationale": "<one sentence grounded in the VLM description and a KAG fact>"}

Figure 15: The reasoning call. The two closed vocabularies are read from the task's store of environmental constraints, so a task that has no insertion phase does not offer one. The rationale field is what forces the label to be traceable to both the visual evidence and a named constraint.

Push-T stores its bounds as typed workspace nodes and its controller as a node in its own right.

```
{
  "id": "ws_tee", "type": "Workspace",
  "label": "Reliable tee init range",
  "properties": {
    "x": [-0.20, 0.20],
    "y": [-0.25, 0.05],
    "z": 0.021
  },
  "id": "ws_tcp", "type": "Workspace",
  "label": "Reliable tcp range",
  "properties": {
    "x": [-0.35, 0.35],
    "y": [-0.35, 0.35],
    "z": [0.02, 0.08]
  },
  "id": "ctrl", "type": "Controller",
  "label": "pd_joint_pos / rel_joint_pos",
  "properties": {
    "policy_action": "7 joint deltas (rel_joint_pos)"
  }
}
```

```
"expert_action": "PPO ->
  joint_delta_pos (same
  7-joint space)"}}
```

The predicate the feasibility loop checks is stored as an implication and is written in the imperative, because it is addressed to the model as much as to the checker.

```
"workspace_constraint": "Every
prescribed config MUST keep tee_xyz
within x[-0.20,0.20] y[-0.25,0.05]
z=0.021 and tcp_xyz within
x[-0.35,0.35] y[-0.35,0.35]
z[0.02,0.08]; out-of-range poses
are dropped (the PPO expert is
unreliable there) and waste the
round."
```

That constraint is also where the Push-T expert's partial competence is encoded. The expert pushes in one rotational direction only, so the configurations it cannot demonstrate

SYSTEM: You are a demonstration coach for an interactive imitation-learning loop. Each round you spend ONE expert demonstration to fix the dominant failure mode. You decide HOW to spend it, grounded in the KAG facts and the per-failure analyses. Reason briefly, then end with EXACTLY two lines: (1) a decision line in the exact required format, and (2) a confidence line 'CONFIDENCE: <integer 0-100> - <one-line rationale>' reporting how confident you are that this demonstration will improve the policy.

USER: (A) SELECT ep<ID> - one recorded failure represents the whole mode. That exact scene is re-run and the expert corrects it on-policy from the divergence point t*. Use when the cluster is TIGHT or one failure clearly dominates.

(B) BRIDGE ep<ID>,ep<ID> - no single failure covers the mode. Prescribe ONE new object placement in the MIDDLE GROUND between 2-3 cited failures (e.g. failures at (1,1) and (5,5) -> a demo near (3,3)); the expert demonstrates from there. Use when the members are geometrically SPREAD but share a root cause.

Figure 16: The prescription call. Arm (B) is offered only where the task's constraint store declares it applicable, so the two arms are read from the store rather than hard-coded. Wipe declares itself targeted-only and the bridging option is omitted from its prompt.

are excluded before a prescription is issued rather than discovered after the expert has failed on one.

Door's constraint is tighter by an order of magnitude, and the numbers are a padded empirical measurement of the simulator's own reset sampler, so that a prescribed configuration cannot leave the task's native reset distribution.

```
{
  "id": "ws_door",
  "type": "Workspace",
  "label": "Reliable door-frame range",
  "properties": {
    "x": [-0.135, -0.108],
    "y": [-0.366, -0.340],
    "z": 1.10,
    "yaw_rad": [-1.82, -1.57]}},
{
  "id": "succ",
  "type": "SuccessCondition",
  "label": "Door open",
  "properties": {
    "metric": "hinge_qpos > 0.3 rad",
    "info_key": "success"}}
```

For a discrete task the environmental constraint is not a bounding box but a reachability predicate, and GridWorld's store states it as one. A prescribed layout must place the start cell, the goal cell and the three obstacle cells as distinct in-grid cells, with a minimum separation between start and goal and a breadth-first path from one to the other that avoids the obstacles. A layout that fails the predicate is rejected before it reaches the expert.

Wipe shows the store doing something a bounding box cannot do at all. Its document carries an implication that removes an entire arm of the prescription from the task.

```
"select_only": "Wipe randomizes a
whole marker PATH, not a single
object pose, so BRIDGE is
```

```
infeasible - always choose SELECT
of the most representative failed
episode."
```

The planner reads that implication structurally and the prompt omits the bridging option. The store therefore does two jobs. It constrains where a demonstration may be placed, and it determines which prescription arms exist for a task at all. Both jobs are done by knowledge that is written down and checkable, which is the sense in which the framework's model of the environment is explicit rather than implicit in a network's weights.

M. Qualitative Examples and Failure Cases

The modes the partition discovers. Figure 17 shows the three failure modes discovered on Push-T (image) with three sampled members of each. The modes are behaviourally distinct: the block is brought to the goal region but left almost fully inverted; the arm never establishes a working contact; the block is pushed but abandoned at a moderate orientation error and far from the end-effector. The clusters are found from geometry alone and the naming pipeline supplies their labels from the task store's own vocabulary. Three is not the number of failure modes on these tasks. It is the count most often selected when the silhouette criterion runs, and the count varies by round, as study A15 reports.

A round, end to end. The cited failures are handed to the prescription model one line each and the model answers in two lines. The round below is taken verbatim from a Wipe log.

```
DOMINANT FAILURE CLUSTER
(members with their VLM+analysis
findings):
- ep3000001:
```



```

    object_xy=(0.221,-0.027)
    progress=74/500
    peak_loss=0.0434
    root_cause=contact_instability
    phase=transport
- ep3000003:
    object_xy=(0.165,-0.068)
    progress=146/500
    peak_loss=0.0181
    root_cause=contact_instability
    phase=transport
- ep3000006:
    object_xy=(0.123,-0.157)
    progress=234/500
    peak_loss=0.0186
    root_cause=pose_mismatch
    phase=placement
===== RESPONSE =====
SELECT ep3000001
CONFIDENCE: 85 - Ep3000001's
contact_instability root cause (higher
peak_loss) at early progress best
represents the dominant failure mode,
ensuring the expert demonstration
directly addresses unstable wiping
pressure causing missed coverage.

```

An impure cluster, from the inside. That record also shows the imperfection of a geometric partition. Two of the three cited members carry the label `contact_instability` and the third carries `pose_mismatch`, so this mode has a purity of two thirds. The purity measured across the ablation settings, 0.84 to 0.91 in study A14, is an average over instances like this one, and the failure case is the one A14 describes in aggregate: the third episode is at a similar place in the workspace and arrived there by a different process, and geometry cannot separate the two. The prescription in this round is unaffected, because the selected episode is a member of the majority cause, but nothing in the framework guarantees that.

The failure cases the framework has. Four are visible in the record above and in the studies. A cluster can carry more than one root cause, which A14 measures and the example illustrates. A round can fail to produce a feasible prescription within $J_{\max}$ attempts and fall back on the nearest untried failure, which happens in 27 to 35 per cent of rounds once the constraint store is removed and which A6 prices. A round can have too few failures to partition at all, which happens in 15 to 21 per cent of rounds and concentrates at the end of the budget, which A15 and A17 measure. And a task can begin the budget above the band in which allocation matters, which is what Lift does and why Lift is uninformative about any mechanism.

N. Extended Limitations and Outstanding Items

The main paper states the limitations that bound its claim. This section states them in full, ordered by how much they

constrain the claim, and then lists the items that are open.

The selector reasons about one round and knows nothing about the dataset. The selector consumes the current policy’s failures, partitions them into failure modes, and prescribes one demonstration. Its only representation of history is the recency-discounted penalty applied to the clusters already corrected, which records *where* corrections have been placed in the descriptor space and holds no representation of *what the training set contains*. The consequence bounds the framework’s central objective. The selector can determine that the policy failed at a particular configuration and that the root cause is, say, a grasp failure. It cannot determine that the training set already holds six demonstrations of that cause, and that the failure is therefore a symptom of underfitting and not of under-coverage. Under a restricted budget, a demonstration spent re-teaching material the dataset already contains is a demonstration lost, which is the opposite of what the budget exists to buy. Nothing in the current instantiation prevents that expenditure and nothing in it measures how often it occurs. The gap is not closed by the cluster memory even in principle: a memory indexed on geometry answers “have we placed a demonstration near here?” and not “does the dataset already contain this behaviour?”, and the two questions come apart wherever geometry does not determine cause, which A14 measures directly. Retrieval methods index a corpus by embedding distance and select from it (Memmel et al. 2025); others reweight a corpus by domain (Hejna et al. 2024) or score it by quality (Belkhale, Cui, and Sadigh 2023). This framework has no such index, and its selector was never given one.

The descriptor is designed by hand, and geometry recovers cause only where configuration determines cause. A10 defends the descriptor’s *width* on a criterion independent of the reported success rates, and what it establishes is that within this family of geometric features six dimensions is the right number. It says nothing about whether a learned descriptor, of the kind that frozen visual encoders supply for the image-modality policies (Nair et al. 2022), would separate failure modes better. That experiment has not been run. A14 locates the descriptor’s ceiling, and purity is measured against the prescription model’s own labels and covaries with silhouette, so neither measurement audits the other. A human-labelled root-cause set is the measurement that is missing.

The cluster memory is worth about three quarters of a point, and how much depends on the task. No study establishes when a task should switch it on. The three settings disagree by a factor of three, three settings cannot support a rule, and the rule is not offered.

Each language-model component is worth about one and a third points, for a structural reason. The comparison that sharpens the point is the grounding knockout: removing the prescription model costs 1.33 points, and removing the environmental constraints against which its proposals are verified costs 2.37 and drives the fallback rate to between 27 and 35 per cent of rounds. Two readings are admissible, that language models add little to demonstration selection,

or that the language model here was given too little to reason over. The evidence supports the second. The literature says why this should be expected, since vision-language models are reliable at naming a cause given structured evidence and unreliable at metric geometry from pixels alone (Chen et al. 2024a; Fu et al. 2024). The framework hands the model a low-dimensional geometric descriptor and a constraint store for exactly that reason, and it then does not hand it anything else.

The allocation machinery is active early and idle late. Failures per round fall from forty-two to two, and below four remaining failures the sweep is skipped. Across the three ablation settings, 15 to 21 per cent of all rounds skip clustering. The allocation account is an account of the early and middle rounds, and the failure-count curve is instrumented on one setting only.

The reasoning pipeline costs seconds and tokens. The add-on the baseline never pays is 63 seconds per round at the cheapest setting and 1,232 at the most expensive, at 9,560 to 82,116 tokens. The round ratio of 1.13 to 2.75 is the smaller of the two facts, because the retrain and the evaluation that both arms pay dilute it. A ratio near one measures the retrain’s expense, not the pipeline’s cheapness.

The experiments are in simulation and no expert is a person. Every result comes from simulation with an expert that answers instantly and correctly, and charges the same for any prescription. The uniformity assumption is what allows the budget to be counted in demonstrations. Outside simulation the budget is a person’s time, demonstrations differ by an order of magnitude in what they cost to produce, and human demonstrators are not uniformly correct (Belkhale, Cui, and Sadigh 2023; Khazatsky et al. 2024; Mandlekar et al. 2020, 2018). The second half of the information-gain argument fails the moment the demonstrator is a person, because high pre-retrain loss identifies novel data only if invalid demonstrations are ruled out by construction.

Outstanding items. Five are open and none is resolved by an assumption. The bridging discrepancy of A16, in which 24 per cent of GridWorld prescriptions are marked bridged on a discrete grid, awaits an inspection of the prescription logs. The decomposition of the constraint store, which A6 does not perform, leaves it unknown whether the workspace bounds alone would recover most of the damage. The cleaner variant of A3, which keeps the descriptor and replaces agglomerative clustering with a random partition into k groups, was not run, so A3 knocks out the allocation stack rather than the partition in isolation. The solvability screen is not exercised in any reported run and is not ablated anywhere. And the failure-count instrumentation of A17 exists on one setting, so the front-loading behaviour is not claimed as general.

Table 1: Every hyperparameter of the reported runs. The cluster-memory weight, recency discount and bandwidth are held at one global setting per task and the memory is switched on in every run reported in the main paper; study A1 in Section F prices it. Study A11 in Section G sweeps the budget over 10, 20 and 40.

Quantity	Value
Budget B	20 demonstrations
Demonstrations per round D	1
Rounds per run	$B/D = 20$
Seeds, GridWorld	9
Seeds, robot tasks	5
Initial demonstrations $ \mathcal{D}_0 $	20 GridWorld, 20 Push-T, 12 Wipe, 8 Lift, 4 Door
Retraining, GridWorld	every round, from scratch
Retraining, robot tasks	every fourth demonstration
Descriptor width	6 (study A10)
Standardisation	zero mean, unit variance
Partition	agglomerative, Ward linkage
Cluster count k	$\arg \max$ mean silhouette
Sweep range	$[2, k_{\max}]$, with $k_{\max} = \max(2, \min(6, N - 1))$
Minimum failures to cluster	4
Context-set cap κ	3 (study A13)
Diversity fill	farthest-point sampling
Re-prescription limit $J_{\max}$	5
Fallback rule	nearest untried failure
Threshold η	quantile of the training-loss distribution, recalibrated at each retrain
Run length K	consecutive steps above η
Vision-language model	Qwen3-VL (Bai et al. 2025)
Reasoning model	Qwen3 (Yang et al. 2025)
Prescription model	Qwen3 (Yang et al. 2025)
Frames per cited failure	3 (start, t^* , end)
Image-branch encoder	R3M (Nair et al. 2022)

Table 2: The six components of the geometric descriptor ϕ per task. The descriptor is computed from privileged simulator state at the flagged step and is identical under both observation modalities. No output of any foundation model enters it.

Task	Components of ϕ
GridWorld	agent cell (2), signed offset to goal (2), progress, Manhattan distance to goal
Push-T	block planar position (2), $\sin \psi$, $\cos \psi$, progress, end-effector-to-block distance
Lift	cube planar position (2), progress, gripper-to-cube distance, gripper height, grasp indicator
Door	door-frame position (2), frame yaw, normalised hinge angle, end-effector-to-handle distance, progress
Wipe	remaining-dirt centroid (2), proportion wiped, end-effector-to-centroid distance, markers remaining, progress

Table 3: Every knockout, on the three ablation settings, ordered by mean damage. Values are the change in final success rate in points against full DISEIL. Mean margin retained is the mean over the three settings of (ablated – best baseline)/(full – best baseline), as a percentage. A negative value means the ablated system finished, on average, beneath the baseline it was built to beat. GW is GridWorld (image), PT is Push-T (state) and Dr is Door (image).

Study	Component knocked out	Δ success rate (points)			Mean	Mean margin retained (%)
		GW	PT	Dr		
A3	the clustering step	−2.2	−4.1	−6.8	−4.37	−53.2
A8	the prescription, fallback rule promoted	−1.8	−3.6	−4.4	−3.27	−20.6
A6	the environmental constraints	−1.5	−2.7	−2.9	−2.37	10.3
A4	the prescription model	−0.5	−1.9	−1.6	−1.33	48.6
A5	the vision-language model	−0.6	−2.0	−1.4	−1.33	47.0
A7	bridging placement	−1.3	−1.1	−1.4	−1.27	51.4
A1	the cluster memory	−0.6	−0.4	−1.2	−0.73	75.1

Table 4: The allocation ladder in absolute terms: final held-out success rate (per cent) on the three ablation settings. A2 replaces allocation with a uniform draw over recorded failures, A8 promotes the deterministic fallback rule to the whole method, and A3 keeps the loss signal and removes the mode structure. GW is GridWorld (image), PT is Push-T (state) and Dr is Door (image).

Arm	GW	PT	Dr
A2 uniform-random allocation	89.1	82.3	80.0
A8 fallback rule only	89.5	92.5	84.2
A3 greedy worst-loss	89.1	92.0	81.8
full DISEIL	91.3	96.1	88.6

Table 5: Mean silhouette of the failure clusters against the width of the geometric descriptor (study A10), averaged over the three ablation settings. Six dimensions is the highest-scoring variant in each of the three settings individually. Silhouette scores geometric separation only and is independent of the success rate.

Descriptor dimensions	Mean silhouette
2	0.373
4	0.507
5	0.557
6 (the framework’s setting)	0.593
8	0.550
10	0.490
12	0.423

Table 6: The composition and the size of the context set (studies A9 and A13), as the mean cost in success-rate points over the three ablation settings against a full-system reference of 92.0. The single-citation arm was not run because bridging requires at least two cited failures to define a placement, so it is confounded by construction and is not a null result.

Arm	Cost (pts)
<i>A9, composition; three citations</i>	
Drop the forced representative	−3.2
Drop the diversity fill	−3.2
Drop the worst-loss seed	−3.27
Three episodes drawn at random	−3.6
<i>A13, number of citations</i>	
Top three by plain peak-loss rank	−2.13
Two citations	−1.93
Five citations	no gain
One citation	not run

Table 7: Root-cause label purity per geometric cluster (study A14). Mean cluster purity is the fraction of a geometric cluster’s failures that share the dominant root cause assigned by the prescription model. Mean root causes per cluster counts the distinct root causes present in a cluster. Purity is scored against the prescription model’s own labels, so it records agreement between two components of the same system and not agreement with ground truth.

Setting	Purity	Causes	Silhouette
GridWorld 5x5 (image)	0.89	1.62	0.58
Push-T (state)	0.91	1.35	0.64
Door (image)	0.84	1.86	0.56

Table 8: Per-round wall-clock and token cost, DISEIL against SafeDagger (study A18). *Shared* is the part of the round that both arms pay: a from-scratch policy retrain and a held-out evaluation. *Add-on* is every DISEIL-specific stage of the round, which is the failure-screening rollout together with clustering, the vision-language call, the reasoning call, the prescription and the feasibility check, less the query-gate rollout the baseline runs in its place. *Tokens* is the sum of the vision-language and language-model tokens drawn in the round. Token counts are not comparable across rows, because the settings were served by different backends and the hidden reasoning tokens are read directly from the usage record on some and recovered from the billed completion length on others.

Protocol	Setting	Baseline (s)	DISEIL (s)	Shared (s)	Add-on (s)	Tokens
P1	Door (state)	737.0	1,054.0	783.0	+270.0	11,511
P1	Door (image)	1,247.0	1,474.0	1,180.0	+293.0	11,379
P1	Wipe (image)	1,468.0	2,195.0	1,491.0	+700.0	9,560
P1	Push-T (image)	688.0	1,891.0	652.5	+1,232.1	82,116
P1	GridWorld (image)	54.6	118.0	51.1	+62.6	9,735
P5	Door (state)	532.6 ± 145.5	782.6 ± 183.5	547.8	+232.8	10,929
P5	Door (image)	$1,034.2 \pm 150.2$	$1,168.8 \pm 194.8$	901.6	+266.2	10,787
P5	Wipe (image)	$1,303.3 \pm 144.1$	$1,991.0 \pm 179.2$	1,332.7	+655.7	10,118

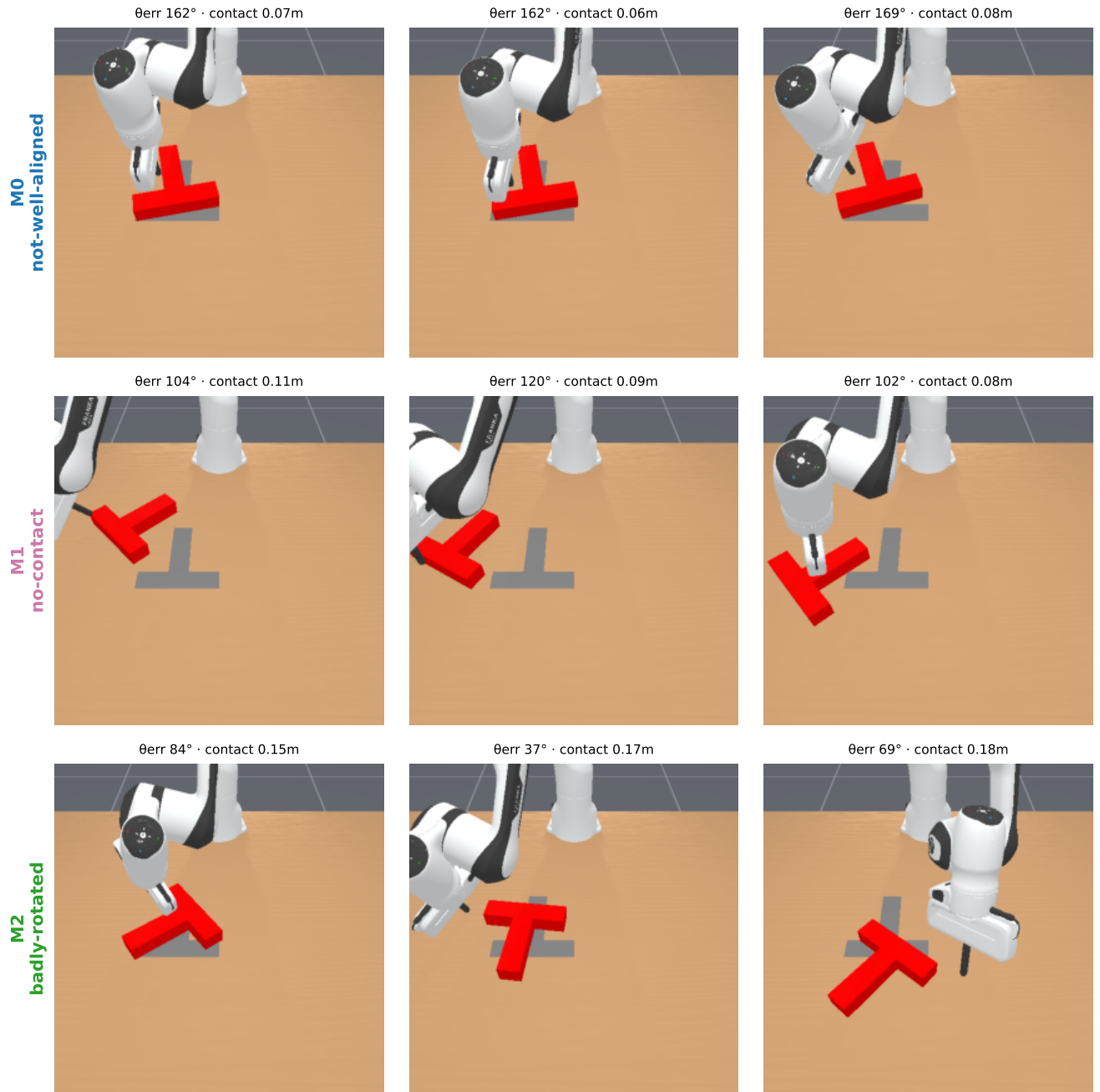


Figure 17: Failure modes discovered on Push-T (image). Each row is one geometric cluster of the round’s recorded failures, with three sampled members. Labels are assigned by the naming pipeline from the task store’s vocabulary. The figure illustrates the object the framework allocates over; the evidence that the partition buys the result is study A3.

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